

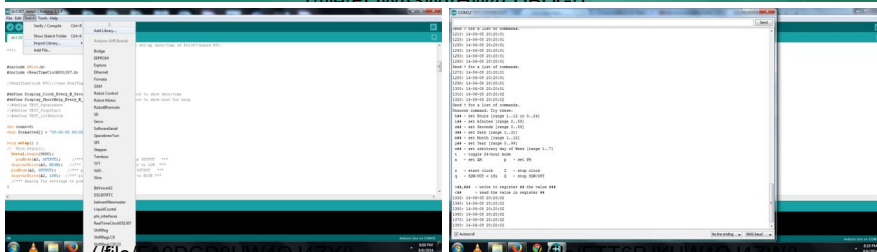
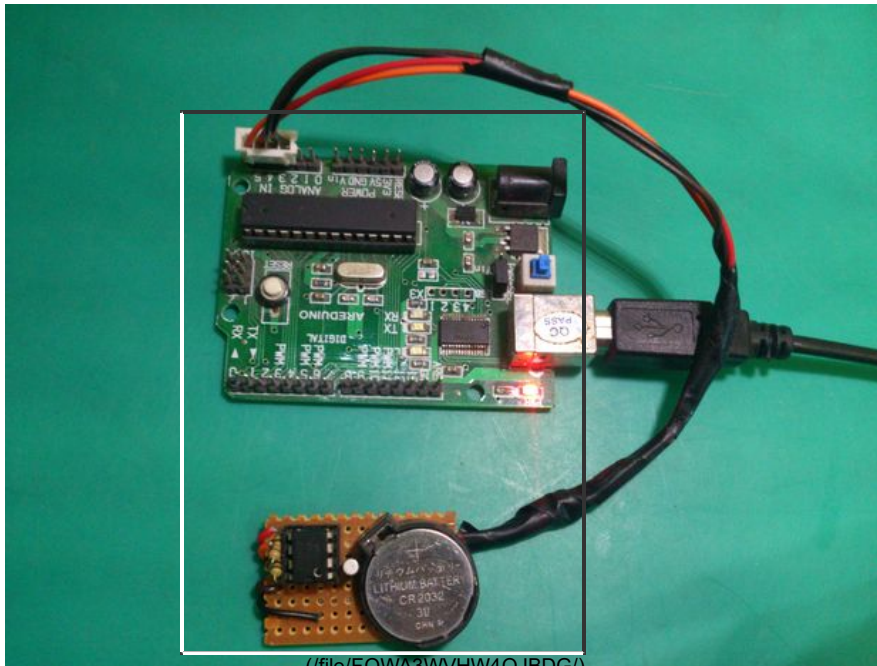


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Step 4: Checking and Setting The Module



Once you are done assembling the module. Connect the module to the Arduino, so that the pins are inserted properly into the Arduino. The program for testing the module is given below. The code constantly updates the time through the serial port.

Arduino Real Time Clock (DS1307) by Naren_Murali (/member/Naren_Murali/)

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Importing The Libraries. (/id/Arduino-Real-Time-Clock-DS1307/?ALLSTEPS) 7 Steps (/id/Arduino-Real-Time-Clock-DS1307/step/5/About-the-library/)

- Download the library "**RealTimeClockDS1307**" and save it in the desktop.
- Open the Arduino and go to **Sketch => Import Library => Add Library**.
- Then select the library saved in the desktop and click **add**.
- Now **paste the sample code** given below and click **compile**. If it compiles successfully then skip the remaining steps.
- If it doesn't import the second library "**Wire**" and repeat the same procedure and compile the code and it will work.

Code Working:

The code is written by "David H. Brown" I merely use it to give you an introduction to the DS1307. Anyway the Arduino communicates with the RTC Module and updates the time for every second through the **Serial Monitor**. The

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Naren_Murali (/member/Naren_Murali/) (https://www.facebook.com/nare)

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40

Bio: I am currently doing my BE in mechanical engineering but im more interested in physics and robotics and such stuff. I am new here hope ... More » (/member/Naren_Murali/)

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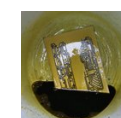
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Arduino Clock - Seven Segment + RTC DS1307 (/id/Arduino-Clock-Seven-Segment-RTC-DS1307/)

creator has given us a option to set the time when we send the command "?". The following menu appears.

Try these:

```
h## - set Hours [range 1..12 or 0..24]
i## - set mInutes [range 0..59]
s## - set Seconds [range 0..59]
d## - set Date [range 1..31]
m## - set Month [range 1..12]
y## - set Year [range 0..99]
w## - set arbitrary day of Week [range 1..7]
t - toggle 24-hour mode
a - set AM p - set PM

z - start clock Z - stop clock
q - SQW/OUT = 1Hz Q - stop SQW/OUT
```

This code will help you to set the time as well as check if the module's backup battery supply works properly. In the next tutorial I will show you how to setup a LCD Display and display the time from the RTC module. Here is the code for testing the module and an attached file of the same.

```
//RealTimeClock RTC;//=new RealTimeClock();

#define Display_Clock_Every_N_Seconds 10 // n.secs to show date/time
#define Display_ShortHelp_Every_N_Seconds 60 // n.secs to show hint for help
//define TEST_Squarewave
//define TEST_StopStart
//define TEST_1224Switch

int count=0;
char formatted[] = "00-00-00 00:00:00x";

void setup() {
// Wire.begin();
Serial.begin(9600);
pinMode(A3, OUTPUT); //*** pin 16 (Analog pin 2) as OUTPUT ***
digitalWrite(A3, HIGH); //*** pin 16 (Analog pin 2) set to LOW ***
pinMode(A2, OUTPUT); //*** pin 17 (Analog pin 3) as OUTPUT ***
digitalWrite(A2, LOW); //*** pin 17 (Analog pin 3) set to HIGH ***
//*** Analog Pin settings to power RTC module ***
}

void loop() {
if(Serial.available())
{
```



RealTimeClockDS1307.zip

(/files/orig/FC2/O0HR/HW4OIBWY/FC2O0HRHW4OIBWY.zip)



Wire.zip (/files/orig/FGK/61VI/HW4OIBX0/FGK61VIHW4OIBX0.zip)



ds1307_sketch.ino (/files/orig/FPM/LN51/HW4OJBQQ/FPMLN51HW4OJBQQ.ino)



Assemble an Adafruit DS1307 Real Time Clock Kit

Build yourself a clock and thermometer (/id/Build-yourself-a-clock-and-thermometer/)

Arduino Nano: DS1307 Real Time Clock(RTC) with Visuino (/id/Arduino-Nano-DS1307-Real-Time-ClockRTC-Visuino/)

Arduino Nano with DHT11, RTC DS1307 and LCD 16x2 (/id/Arduino-Nano-With-DHT11-RTC-DS1307-and-LCD-16x2/)

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I Made it!

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barancanaydin (/member/barancanaydin) made it!

a year ago

Reply

```
#include <DS1307RTC.h>
#include <LiquidCrystal.h>
#include <Wire.h>
#include <Time.h>
LiquidCrystal lcd(12,11,10,7,6,5,4);
void setup() {
  Serial.begin(9600);
  while (!Serial) ; // wait for serial
  delay(200);
  Serial.println("DS1307RTC Read Test");
  Serial.println("-----");
}
void loop() {
  tmElements_t tm;
  if (RTC.read(tm)) {
    Serial.print("Ok, Time = ");
    print2digits(tm.Hour);
    Serial.write(':');
    print2digits(tm.Minute);
    Serial.write(':');
    print2digits(tm.Second);
    Serial.print(", Date (D/M/Y) = ");
    Serial.print(tm.Day);
    Serial.write('/');
    Serial.print(tm.Month);
    Serial.write('/');
    Serial.print(tmYearToCalendar(tm.Year)-2000);
    Serial.println();
    lcd.begin(8,2); // columns, rows. use 16,2 for a 16x2 LCD, etc.
    lcd.clear(); // start with a blank screen
    lcd.setCursor(0,0);
    lcd.print(tm.Hour);
    lcd.print(":");
    lcd.print(tm.Minute);
    lcd.print(":");
    lcd.print(tm.Second); // change this text to whatever you like. keep it clean.
    lcd.setCursor(0,1); // set cursor to column 0, row 1
    lcd.print(tm.Day);
    lcd.print("/");
    lcd.print(tm.Month);
```

```

lcd.print("/");
lcd.print(tmYearToCalendar(tm.Year)-2000);
} else {
if (RTC.chipPresent()) {
Serial.println("The DS1307 is stopped. Please run the SetTime");
Serial.println("example to initialize the time and begin running.");
Serial.println();
} else {
Serial.println("DS1307 read error! Please check the circuitry.");
Serial.println();
}
delay(9000);
}
delay(1000);
}
void print2digits(int number) {
if (number >= 0 && number < 10) {
Serial.write('0');
}
Serial.print(number);
}
}

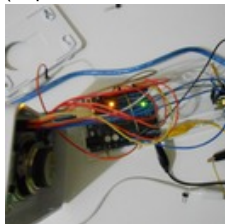
```



(<http://cdn.instructables.com/F5X/ZW7B/I9P1UN7A/F5XZW7BI9P1UN7A.LARGE.jpg>)



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krecik (/member/krecik)

a year ago

Reply

Connect the oscillator's case do GND, it can gain accuracy :).



Naren_Murali (/member/Naren_Murali) (author) ▶ **krecik** (/member/krecik)

a year ago

Reply

Thanks for that!!! I forgot about grounding the oscillator's case!!!



tobyvanreenen (/member/tobyvanreenen) made it!

a year ago

Reply

I used the RTC control sketch from Vaughters and found it to be useful for inclusion in my own projects. I've dumped it below with some of my own tiny additions.

The image attached is of a stripboard version

/* RTC Control v1.001

* by <<http://www.combustory.com>> John Vaughters

*

* THIS CODE IS FREE FOR ANYTHING - There is no Rocket Science here. No need to create some long GPL statement.

*

* Credit to:

* Maurice Ribble - <http://www.glacialwanderer.com/hobbyrobotics>
(<http://www.glacialwanderer.com/hobbyrobotics>) for RTC DS1307 code

* BB Riley - Underhill Center, VT <brianbr@wulfdcn.org> For simplification of the Day of Week and month

* and updating to Arduino 1.0

* peep rada - from Arduino Forum - Found that only 32 registers per I2C connection was possible

*

* With this code you can set the date/time, retrieve the date/time and use the extra memory of an RTC DS1307 chip.

* The program also sets all the extra memory space to 0xff.

* Serial Communication method with the Arduino that utilizes a leading CHAR for each command described below.

*

FOR DIY RTC: <http://www.instructables.com/id/Arduino-Real-Time...>
(<http://www.instructables.com/id/Arduino-Real-Time-Clock-DS1307/>)

*****Connections*****!!

RTC ---- UNO V3 ---- NANO V3.0G

5V -- 5V -- 5V

GND -- GND -- GND

SCL -- A5 -- A3

SDA -- A4 -- A2

****Commands:*****!!

* T(00-59)(00-59)(00-23)(1-7)(01-31)(01-12)(00-99) - T(sec)(min)(hour)
(dayOfWeek)(dayOfMonth)(month)(year) -

* T - Sets the date of the RTC DS1307 Chip.

* Example to set the time for 25-Jan-2012 @ 19:57:11 for the 4 day of the week, use this command - T1157194250112

* Q(1-2) - (Q1) Memory initialization (Q2) RTC - Memory Dump

* R - Read/display the time, day and date

*

* _____

* Notes:

* Version 1.0

* Moving this code to Version 1.0 because this code has been updated to Arduino v1.0 and the features have

* been well tested and improved in a collaborative effort.

* - Fixed the issue of not being able to access all the registers - JWV

* - Added initialization for all non-time registers - JWV

* - Added Dump of all 64 registers - JWV

- * - Some Date/Time reformatting and cleanup of display, added Day/Month texts - BBR
- * - Made compatible with Arduino 1.0 - BBR
- * - Added Rr command for reading date/time - BBR
- * - Made commands case insensitive - BBR
- * - Create #define variables to support pre Arduino v1.0 - JWV
- * Version 0.01

* Initial code with basics of setting time and the first 37 registers and dumping the first 32 registers.

* The code was based on Maurice Ribble's original code.

*

*/

```
#include "Wire.h"
```

```
#define DS1307_I2C_ADDRESS 0x68 // This is the I2C address
```

```
// Arduino version compatibility Pre-Compiler Directives
```

```
#if defined(ARDUINO) && ARDUINO >= 100 // Arduino v1.0 and newer
```

```
#define I2C_WRITE Wire.write
```

```
#define I2C_READ Wire.read
```

```
#else // Arduino Prior to v1.0
```

```
#define I2C_WRITE Wire.send
```

```
#define I2C_READ Wire.receive
```

```
#endif
```

```
// Global Variables
```

```
int command = 0; // This is the command char, in ascii form, sent from the serial port
```

```
int i;
```

```
long previousMillis = 0; // will store last time Temp was updated
```

```
byte second, minute, hour, dayOfWeek, dayOfMonth, month, year;
```

```
byte test;
```

```
byte zero;
```

```
char *Day[] = {"", "Sun", "Mon", "Tue", "Wed", "Thu", "Fri", "Sat"};
```

```
char *Mon[] =
```

```
{"", "Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"};
```

```
// Convert normal decimal numbers to binary coded decimal
```

```
byte decToBcd(byte val)
```

```
{
```

```
return ( (val/10*16) + (val%10) );
```

```
}
```

```
// Convert binary coded decimal to normal decimal numbers
```

```
byte bcdToDec(byte val)
```

```
{
```

```
return ( (val/16*10) + (val%16) );
```

```
}
```

```
// 1) Sets the date and time on the ds1307
```

```
// 2) Starts the clock
```

```
// 3) Sets hour mode to 24 hour clock
```

```
// Assumes you're passing in valid numbers, Probably need to put in checks for valid numbers.
```

```
void setDateDs1307()
```

```
{
```

```

second = (byte) ((Serial.read() - 48) * 10 + (Serial.read() - 48)); // Use of (byte)
type casting and ascii math to achieve result.

minute = (byte) ((Serial.read() - 48) * 10 + (Serial.read() - 48));

hour = (byte) ((Serial.read() - 48) * 10 + (Serial.read() - 48));

dayOfWeek = (byte) (Serial.read() - 48);

dayOfMonth = (byte) ((Serial.read() - 48) * 10 + (Serial.read() - 48));

month = (byte) ((Serial.read() - 48) * 10 + (Serial.read() - 48));

year = (byte) ((Serial.read() - 48) * 10 + (Serial.read() - 48));

Wire.beginTransmission(DS1307_I2C_ADDRESS);

I2C_WRITE(zero);

I2C_WRITE(decToBcd(second) & 0x7f); // 0 to bit 7 starts the clock

I2C_WRITE(decToBcd(minute));

I2C_WRITE(decToBcd(hour)); // If you want 12 hour am/pm you need to set
// bit 6 (also need to change readDateDs1307)

I2C_WRITE(decToBcd(dayOfWeek));

I2C_WRITE(decToBcd(dayOfMonth));

I2C_WRITE(decToBcd(month));

I2C_WRITE(decToBcd(year));

Wire.endTransmission();

}

// Gets the date and time from the ds1307 and prints result

void getDateDs1307()
{
// Reset the register pointer
Wire.beginTransmission(DS1307_I2C_ADDRESS);

I2C_WRITE(zero);

Wire.endTransmission();

Wire.requestFrom(DS1307_I2C_ADDRESS, 7);

// A few of these need masks because certain bits are control bits

second = bcdToDec(I2C_READ() & 0x7f);

minute = bcdToDec(I2C_READ());

hour = bcdToDec(I2C_READ() & 0x3f); // Need to change this if 12 hour am/pm

dayOfWeek = bcdToDec(I2C_READ());

dayOfMonth = bcdToDec(I2C_READ());

month = bcdToDec(I2C_READ());

year = bcdToDec(I2C_READ());

if (hour < 10)

Serial.print("0");

Serial.print(hour, DEC);

Serial.print(":");

if (minute < 10)

Serial.print("0");

Serial.print(minute, DEC);

Serial.print(":");

if (second < 10)

Serial.print("0");

Serial.print(second, DEC);

Serial.print(" ");

Serial.print(Day[dayOfWeek]);

```



```

Serial.print(" ");
Serial.print(dayOfMonth, DEC);
Serial.print(" ");
Serial.print(Mon[month]);
Serial.print(" 20");
if (year < 10)
Serial.print("0");
Serial.println(year, DEC);
}

void setup() {
Wire.begin();
Serial.begin(57600);
zero=0x00;
Serial.println("Ready");
}

void loop() {
if (Serial.available()) { // Look for char in serial queue and process if found
command = Serial.read();
if (command == 84 || command == 116) { //If command = "Tt" Set Date
setDateDs1307();
getDateDs1307();
Serial.println(" ");
}
else if (command == 82 || command == 114) { //If command = "Rr" Read Date ...
BBR
getDateDs1307();
Serial.println(" ");
}
else if (command == 81 || command == 113) { //If command = "Qq" RTC1307
Memory Functions
delay(100);
if (Serial.available()) {
command = Serial.read();
if (command == 49) { //If command = "1" RTC1307 Initialize Memory - All Data
will be set to
// 255 (0xff). Therefore 255 or 0 will be an invalid value.
Wire.beginTransaction(DS1307_I2C_ADDRESS); // 255 will be the init value
and 0 will be considered
// an error that occurs when the RTC is in Battery mode.
I2C_WRITE(0x08); // Set the register pointer to be just past the date/time
registers.
for (i = 1; i <= 24; i++) {
I2C_WRITE(0Xff);
delay(10);
}
Wire.endTransmission();
Wire.beginTransaction(DS1307_I2C_ADDRESS);
I2C_WRITE(0x21); // Set the register pointer to 33 for second half of registers.
Only 32 writes per connection allowed.
for (i = 1; i <= 33; i++) {
I2C_WRITE(0Xff);

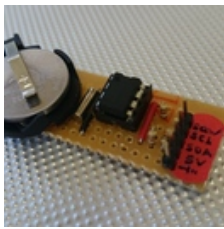
```



```

delay(10);
}
Wire.endTransmission();
getDateDs1307();
Serial.println(": RTC1307 Initialized Memory");
}
else if (command == 50) { //If command = "2" RTC1307 Memory Dump
getDateDs1307();
Serial.println(": RTC 1307 Dump Begin");
Wire.beginTransmission(DS1307_I2C_ADDRESS);
I2C_WRITE(zero);
Wire.endTransmission();
Wire.requestFrom(DS1307_I2C_ADDRESS, 32);
for (i = 0; i <= 31; i++) { //Register 0-31 - only 32 registers allowed per I2C
connection
test = I2C_READ();
Serial.print(i);
Serial.print(": ");
Serial.print(test, DEC);
Serial.print(" : ");
Serial.println(test, HEX);
}
Wire.beginTransmission(DS1307_I2C_ADDRESS);
I2C_WRITE(0x20);
Wire.endTransmission();
Wire.requestFrom(DS1307_I2C_ADDRESS, 32);
for (i = 32; i <= 63; i++) { //Register 32-63 - only 32 registers allowed per I2C
connection
test = I2C_READ();
Serial.print(i);
Serial.print(": ");
Serial.print(test, DEC);
Serial.print(" : ");
Serial.println(test, HEX);
}
Serial.println(" RTC1307 Dump end");
}
}
}
Serial.print("Command: ");
Serial.println(command); // Echo command CHAR in ascii that was sent
}
command = 0; // reset command
delay(100);
}
//*****The End*****

```



(<http://cdn.instructables.com/FQE/3EAK/I376RYEZ/FQE3EAKI376RYEZ.LARGE.jpg>)



Naren_Murali (/member/Naren_Murali) (author) ▶ tobyvanreenen

(/member/tobyvanreenen)

a year ago

Reply

Congrats!!!. The module looks really cool!!



Harshvardhan Singh (/member/Harshvardhan Singh)

a month ago

Reply

How to set time?



mbauld (/member/mbauld)

3 months ago

Reply

Cant you just use a clock signal from the Arduino using time keeping libraries instead of a crystal oscillator??



vyshjyadav (/member/vyshjyadav)

6 months ago

Reply

i need ur help.....

while displaying time using this code it displays as
45:165:85

165/165/165

please help me to solve this,,,im stuck with this,,,



Harshvardhan Singh (/member/Harshvardhan Singh)

7 months ago

Reply

Can we use a ready made RT module. If yes, would the wiring and coding remain same?



(<http://cdn.instructables.com/FSI/JUYJ/IF4N9IDS/FSIJUYJIF4N9IDS.LARGE.jpg>)



Harshvardhan Singh (/member/Harshvardhan Singh)

8 months ago

Reply

Can we use a ready-made DS1307 clock module? If yes, then will the wiring remain same?



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DimasH2 (/member/DimasH2)

8 months ago

[Reply](#)

did u not program A4 and A5?



lamglozekmiles (/member/lamglozekmiles)

9 months ago

[Reply](#)

how can i use the real time clock to switch a 12v dc relay at set time



sauraveee (/member/sauraveee)

9 months ago

[Reply](#)

Everything is nice and great my only problem is that in the first testing ds1307_sketch when i open the serial monitor it just runs and runs i mean i can't see what is written how to stop it means so that stays like in your screen shot of the menu mine is running too fast
my options are selected autscroll no line ending and 9600 baud
thank you!



NafiulT (/member/NafiulT)

10 months ago

[Reply](#)

This video Arduino with RTC Plis like and Subscribe Share

<http://www.youtube.com/watch?v=q3MvkfzpVAY>
(<https://www.youtube.com/watch?v=q3MvkfzpVAY>)



NafiulT (/member/NafiulT)

10 months ago

[Reply](#)



KemalV (/member/KemalV)

10 months ago

[Reply](#)

i had a little trouble with serial communication with ds1307_sketch.ino. Problem was that serial command didnt have enough time to get to buffer. So i added delay of 100ms as a first line to processcommand function.



KalagoniS (/member/KalagoniS)

10 months ago

[Reply](#)

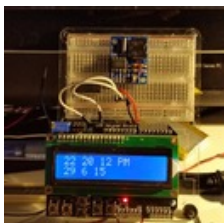


JohnC20 (/member/JohnC20) made it! 

10 months ago

[Reply](#)

Ok I finally got brave and built the DS1307 RTC kit from Adafruit. Since I only had the 16x2 LCD with buttons I had to change the pin assignments. But for a beginner I got it to work the first time. Guess you CAN teach an old dog new tricks.



(<http://cdn.instructables.com/FAT40BY/IBGDATRU/FAT40BYIBGDATRU.LARGE.jpg>)

marsekal (/member/marsekal)



11 months ago [Reply](#)

good job MR, IS THE BEST



IrrigationSystem (/member/IrrigationSystem) a year ago [Reply](#)
Nice and good job ! :) This works only with 5V ? Won't work with 3.3V ?



ArpitPatel (/member/ArpitPatel) a year ago [Reply](#)
Thanks a lot for explaining this project thoroughly. I am electronics students and currently learning arduino. I made this project with the help of your tutorial and it works fine for me. It shows same output as in video and pics. But my question is if i want to make alarm and i add some instructions at the end of program than is it possible to works?

For example :

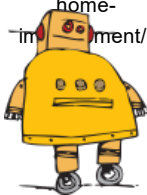
```
if (hours==8 && minutes==15 && seconds<=34)
{
digitalWrite(led,HIGH);
}
else if (hours==13 && minutes==00 && seconds()<=19 )
{
digitalWrite(led,HIGH);
}
```



DangerousTim (/member/DangerousTim) a year ago [Reply](#)
Great tutorial! Very helpful indeed! I'm gonna use this to make my next Arduino project.. Thanks!!!

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